Microeconomics

Lecture 1: Preference and Choice

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Logistics

Textbook: **Microeconomic Theory** by Mas-Colell, Winston and Green, Oxford University Press, 1995

Other useful micro textbooks by Kreps, Varian, Jehle and Reny.

My office hours: 12:00-13:00, Tuesdays, Room 308

Grading: midterm 50%, final 50%, class participation +5%

Course content

Preference and choice

Consumer choice

Classical demand theory

Aggregate demand

Production

Choice under uncertainty (if time allows)

The goal of microeconomics theory

Microeconomic theory concerns the behavior of individual economic actors and the aggregation of their actions in different institutional frameworks.

The basic categories: actors, behavior, institution, and equilibrium

Microeconomics provides tools for all of economics, including empirical economics.

Tools are basic and general and useful for building models.

Question: which parts of a demand model are observed, and which are inferred?

The actors or agents

Consumer: household or individual?

Firm: One approach is assuming a firm as an actor, the other is assuming the firm as an institution in which the behaviors of actors (manager, worker, supplier, and customer) are aggregated.

Behavior

Standard approach: constrained optimization; the actor chooses the option that maximizes an objective function

Objective: utility function, profit; constraint: budget , technological production possibilities set.

Is this type of modeling perfect?

- We are looking for a testable model: what behavior, if observed, would clearly falsify our model? If the model is not falsified by observations, then it is good positive model
- Experimental economics falsifies rationality assumption: in many models, the behavior of one individual is unimportant, but the aggregate behavior of many individual is important.
- Even if there is a systematic violation, the model can give us insight.

Institutional framework

The general nature of options an individual has. The options available to an individual as a function of others' actions.

- competitive markets, auctions, imperfect competition, etc.

Price is impersonal: the individual chooses any bundle they can afford.

Where do prices come from? How are they set? How do they reflect the action of individuals?

Equilibrium analysis

Outcome of the model when

- each individual is doing for itself,
- given the array of actions taken by others,
- and given the institutional framework that defines the options of individuals and links their actions.

The usefulness of falsified models

Knowing what doesn't work is often a good place to figure out what does

Scope, detail, emphasis, and complexity

- What is the appropriate scope of the model. For example, to study general equilibrium, we analyze two competitive markets, why not more?
- The tradeoff between complexity and tractability
- To model consumption, family is treated as one unit, but for labor supply the composition is also important.

Preference and choice

A **preference relation** describes the decision maker's preference among any two hypothetical pairs.

Given this preference, the decision makers will make choices whenever she is offered some set of alternatives to choose from.

Preference is an abstract concept but choice is observable.

When is some choice procedure the by-product of a preference relation?

Can we use observable choices to deduce what is the preference relation guiding those choices?

How to model choice behavior?

There are two distinct approaches to model individuals choice behavior.

In the first approach, we model preferences and impose rationality as a primitive characteristic of individuals.

In the second approach, we directly start by choices and impose weak axiom of revealed preferences parallel to rationality assumption.

This chapter

- preference and the utility representation
- choice structure
- the link between the two alternative approaches

A flash back

Binary relation: a collection of ordered pairs of elements of a set.

$R \subseteq X \times Y$ is a binary relation from X to Y

- We write xRy if $(x, y) \in R$ and *not* xRy if $(x, y) \notin R$.
- When $R \subseteq X \times X$, we say R is a binary relation on X.

Examples

- a function is a binary relation $y = f(x) \iff xRy$,
- weakly greater (smaller) than, $\geq$ ($\leq$),
- strictly greater (smaller) than, $>$ ($<$),
- equal to, $=$.

Some properties of binary relations

A binary relation R on X is:

- **complete** if, for all $x, y \in X$, xRy or yRx;
- **reflexive** if, for all $x \in X, xRx$;
- **irreflexive** if, for all $x \in X$, not xRx;
- **symmetric** if, for all $x, y \in X, xRy$ implies yRx;
- **asymmetric** if, for all $x, y \in X, xRy$ implies not yRx;
- **antisymmetric** if, for all $x, y \in X, xRy$ and yRx imply $x = y$;
- **transitive** if, for all $x, y, z \in X, xRy$ and yRz imply xRz;
- **negatively transitive** if, for all $x, y, z \in X$, not xRy and not yRz imply not xRz.

The objects of choice

X is a set objects to choose.

Usually we assume X as a subset of $\mathbb{R}^k$ where k is the number of consumption alternatives.

$x = (x_1, x_2, \ldots, x_k)$ is a specific bundle of goods.

Ex: tea and coffee. $x = (x_1, x_2)$ represents x_1 cups of tea and x_2 cups of coffee.

Assume y is another bundle of goods. Then, we may ask which one do you prefer?

Time, location, and other relevant circumstances must be included in the description when they affect the alternative.

Preference relations

Preference $\succsim$: a binary relation to compare the pairs of alternatives in X.

$x \succsim y$ means x is at least as good as y.

Strict preference $\succ$: $x \succ y \iff x \succsim y$ and not $y \succsim x$.

Indifference $\sim$: $x \sim y \iff x \succsim y$ and $y \succsim x$.

Indifference is not incomparability: Incomparability means neither $x \succsim y$ nor $y \succsim x$. Completeness excludes this case.

Rational preference

Definition

The preference relation $\succsim$ is **rational** if it is **complete** and **transitive**.

Completeness: an individual can compare all alternatives (even without experience)

Transitivity is the heart of rationality. Preferences are consistent and there is no loop such that $x \succ y \succ z \succ x$.

Rational preference properties

Proposition

If $\succsim$ is rational then:

- *$\succ$ is irreflexive, asymmetric, and transitive;*
- *$\sim$ is reflexive, symmetric, and transitive;*
- *if $x \succ y \succsim z$, then $x \succ z$.*

"Rational" here means consistency of the ordering, not that the person is selfish, informed, or good at arithmetic.

More on rationality

Four possible responses for comparison of x and y

1. x is better than y, not the reverse.
2. y is better than x, not the reverse
3. neither is better
4. x is better than y and y is better than x

Rationality precludes 4 from the answers. Is it always reasonable?

More on rationality

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When the choice is made? compare tea vs. cola

- in summer tea$\succ$cola, in winter cola$\succ$tea.
- transitivity is reasonable at a single point in time.

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Is recognition possible? Shades of gray:

- one grade difference is not recognizable, but a ten-grade is
- so $x_1 \sim x_2 \sim \cdots \sim x_{10}$, but $x_1 \succ x_{10}$.

Validity of rationality

An example by Kahneman and Tversky (1984):

You are about to buy a calculator for \$15 and a stereo for \$125. You are at a store. There is another store 10 minutes away.

- You learn there is a \$5 discount on calculator at the other store. Do you make the trip?
- You learn there is a \$5 discount on stereo at the other store. Do you make the trip?

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Many people say yes to the first and no to the second.

More on rationality

Consider the choice between two vaccination programs

1. saving 400 with certainty vs. saving 600 with probability 2/3;
2. dying 200 with certainty vs. dying 600 with probability 1/3.

More on rationality

Consider the choice between two vaccination programs

1. saving 400 with certainty vs. saving 600 with probability 2/3;
2. dying 200 with certainty vs. dying 600 with probability 1/3.

Medical professionals prefer first in 1 and second in 2, though the outcome is the same.

Framing problem: when items are complex, cognitive process to rank items may be affected by the framing of the problem.

Also see other examples in Mas-colell page 8 (condorcet paradox, change of taste, etc.) and works by Daniel Kahneman & Amos Tversky.

Critiques of rationality assumption

Rationality is probably the weakest part of microeconomics.

- Little psychology.
- Source of preferences or reasons for choices are not explained.
- Consistency in preference or choice is required.

Modern fields (e.g. behavioral economics, behavioral finance, neuroeconomics) have evolved to overcome these weakness using psychological concepts, rather than a simple presumption of rationality.

Learning models, bounded rationality, imitating behavior models, ...

Utility function

Turning preference to numbers using utility function:

Definition

$u : X \to \mathbb{R}$ is a utility function representing preference relation $\succsim$ if,

$$\forall x, y \in X, \qquad x \succsim y \iff u(x) \geq u(y)$$

Preference relation is ordinal, utility is cardinal, so

- utility function of $\succsim$ is not unique, for any strictly increasing function f, $f(u(x))$ is another utility function of $\succsim$.

Utility function and rationality assumption

Proposition

A preference relation $\succsim$ can be represented by a utility function only if it is rational.

Proof.

We show that if u represents $\succsim$, then $\succsim$ is complete and transitive.
$\forall x, y \in X : u(x) \geq u(y)$ or $u(y) \geq u(x) \Rightarrow x \succsim y$ or $y \succsim x$
$x \succsim y, y \succsim z \Rightarrow u(x) \geq u(y), u(y) \geq u(z) \Rightarrow u(x) \geq u(z) \Rightarrow x \succsim z$ □

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Can all rational preferences be represented by a utility function?

Ex: if preferences are like dictionary ordering

We'll see that all "continuous" preferences have a utility representation.

Choice rule

The second approach to model consumer behavior. A choice structure $(\mathfrak{B}, C(.))$ consists of

- $\mathfrak{B}$ a set of subsets of X: if $B \in \mathfrak{B}$ then $B \subset X$.
- $C(.)$ a choice rule: for every budget set B a non-empty set of chosen elements $C(B) \subset B$.

$\mathfrak{B}$ need not to include all possible subsets of X.

$C(B)$ may contain more than one element.

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$C(B)$ may contain more than one element.

Ex: choice between cups of tea and coffee $B = \{(40, 2), (20, 9)\}$.

Consumers might find comparison of the bundles difficult and unable to choose. The assumption of non-emptiness rules this out: $C(B) \neq \emptyset$

But we don't rule out choosing both $C(B) = B$ (indifference between the two bundles)

The weak axiom of revealed preference (WARP)

A reasonable restriction on individuals choice behavior:

Definition

WARP is satisfied if for some $B \in \mathfrak{B}$ with $x, y \in B$, we have $x \in C(B)$, then for any $B' \in \mathfrak{B}$ with $x, y \in B'$ and $y \in C(B')$, we must also have $x \in C(B')$.

If x is chosen once y is available, then it's not possible that y is chosen and x is not when both are available.

Ex: if $C(\{x, y\}) = \{x\}$ then $C(\{x, y, z\}) = \{y\}$ violates WARP.

Why "weak" axiom? you see in the exercises!

Revealed preference relation

Definition

The revealed preference relation $\succsim^*$ is defined by

$$x \succsim^* y \iff \text{there is some } B \in \mathfrak{B} \text{ such that } x, y \in B \text{ and } x \in C(B)$$

$x \succsim^* y$ is read as "x is revealed at least as good as y."
$x \succ^* y$ is read as "x is revealed preferred to y."

WARP: if x is revealed at least as good as y, then y cannot be revealed preferred to x.

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Ex: $X = \{x, y, z\}$, $\mathfrak{B} = \{\{x, y\}, \{x, y, z\}\}$, and two choice rules:
$C_1(\{x, y\}) = \{x\}$, $C_1(\{x, y, z\}) = \{x\}$
$C_2(\{x, y\}) = \{x\}$, $C_2(\{x, y, z\}) = \{x, y\}$
which one does satisfy the WARP?

From preference to choice

Does rational preference lead to choice structure that satisfies WARP?

Define set of acceptable alternative from B as

$$C^*(B, \succsim) = \{x \in B : x \succsim y \text{ for every } y \in B\}$$

- C^* is by definition a subset of B.
- It may contain more than one element.
- If X is finite, C^* is non-empty.
- C^* may be empty, if X is infinite. Consider $X = [0, +\infty)$ and you alway prefer more money. Then, $C^*(A, \succsim) = \emptyset$.
- Here, we only consider cases with non-empty C^*.

From preference to choice

Proposition

Suppose that $\succsim$ is a rational preference relation. Then the choice structure generated by $\succsim$, $(\mathfrak{B}, C^*(.,\succsim))$, satisfies the WARP.

Proof.

$x, y \in B, x \in C^*(B,\succsim) \Rightarrow$ by definition of $C^* : x \succsim y$

$x, y \in B', y \in C^*(B',\succsim) \Rightarrow \forall z \in B' : y \succsim z$

$\Rightarrow \forall x \in B' : \; x \succsim z \Rightarrow x \in C^*(B',\succsim)$ □

Thus, rational preferences suffice the consistency requirement embodied in WARP.

From choice to preference

If choice rule satisfies WARP, is there always a rational preference consistent with it? Maybe.

Definition

Given a choice structure $(\mathfrak{B}, C(.))$, we say the rational preference $\succsim$ **rationalizes** $C(.)$ relative to $\mathfrak{B}$ if $C(B) = C^*(B, \succsim)$

Suppose that $X = \{x, y, z\}$, $\mathfrak{B} = \{\{x, y\}, \{y, z\}, \{x, z\}\}$,
$C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$, $C(\{x, z\}) = \{z\}$
Is WARP satisfied? Can we find a rationalizing preference?

What if we have $\{x, y, z\} \in \mathfrak{B}$? Can WARP then be satisfied?

From choice to preference

It turns out that if $\mathfrak{B}$ includes enough subsets of X, then WARP results in rationality.

Proposition

If $(\mathfrak{B}, C(.))$ is a choice structure such that

1. *the WARP is satisfied, the*
2. *$\mathfrak{B}$ includes all subsets of X of up to three elements,*

then there is a unique rational preference relation $\succsim$ that rationalizes $C(.)$ relative to $\mathfrak{B}$.

If the choice rule is defined on all subsets of X, then WARP is equivalent to rationality.